gradient propagation. After the final block, we apply RMSNorm [113] to bound per-sample feature norms before value heads, preventing out-of-distribution inputs from producing unbounded activations that destabilize bootstrapping.

Pre-activation Batch Normalization. Replay data is collected by a mixture of evolving policies, inducing non-stationary input distributions. Without normalization, feature activations can saturate (e.g., dead ReLUs [1]), degrading gradient flow [13, 54, 43]. We apply batch normalization [33] before each nonlinearity to keep activations well-scaled. We choose batch normalization over layer normalization [5] because it exploits large-batch statistics from diverse replay data, yielding a smoother loss landscape with a lower effective condition number [83, 8, 72].

Cross-Batch Value Prediction. Batch normalization computes statistics per batch, so the predicted Q-values and target Q-values receive different normalization when computed in separate forward passes. Following [8], we concatenate current and next-state transitions into a single batch so that both share the same statistics, ensuring consistency in the Bellman update.

Distributional Critic with Adaptive Reward Scaling. Following [45, 72], we represent the Q-value as a categorical distribution over n_{atom} atoms uniformly spaced on $[G_{\min}, G_{\max}]$. The network predicts atom probabilities and is trained via cross-entropy loss against the projected Bellman target [7]. This distributional formulation smooths the optimization landscape and reduces sensitivity to noisy targets [72].

To keep returns within the distributional critic's fixed support, we normalize rewards directly rather than centering returns [65] or scaling losses [84]. We track the running discounted return variance $\sigma_{t,G}^2$ and maximum magnitude $G_{t,\max}$, and scale as:

$$\bar{r}_t = \frac{r_t}{\max\left(\sqrt{\sigma_{t,G}^2 + \epsilon},\ G_{t,\max}/G_{\max}\right)}. \tag{6}$$

This bounds effective returns while maintaining a consistent scale throughout training.

Weight Normalization. Uncontrolled weight growth increases Q-value variance and amplifies estimation errors under bootstrapping [55]. After each gradient step, we project each weight vector onto the unit-norm sphere [102, 53, 45, 71, 72] and each normalization parameter vector (γ, β) to norm $\sqrt{d}$. This constrains the network to encode information through direction rather than scale.

4.3 Exploration

Off-policy RL can decouple data collection from policy optimization, allowing exploration strategies that pursue broad state-action coverage independently of the current policy. FlashSAC employs two complementary mechanisms.

Unified Entropy Target. Maximum-entropy RL with automatic temperature tuning [22] encourages sustained exploration, but requires specifying a target entropy. Standard practice sets this target per task, which is impractical across embodiments with varying action dimensions. We instead parameterize the target entropy via a fixed action standard deviation σ_{tgt}. For a Gaussian policy with diagonal covariance, this gives:

$$\bar{\mathcal{H}} = \tfrac{1}{2}|\mathcal{A}| \log\left(2\pi e\, \sigma_{\text{tgt}}^2\right), \tag{7}$$

which scales linearly with action dimension, ensuring consistent exploration across embodiments without per-task tuning. We set $\sigma_{\text{tgt}} = 0.15$ in all experiments.

Noise Repetition. Temporally correlated action noise is commonly used to improve exploration in sparse-reward settings, with pink noise [14] and Ornstein–Uhlenbeck noise [29] being widely used. However, these methods are ill-suited to massively parallel simulations, as they require per-environment correlated-noise processes, which incur substantial computational and memory overhead.

We propose *Noise Repetition*, a lightweight alternative that induces temporal correlation using minimal local state. At each repetition interval, a noise vector $\epsilon \sim \mathcal{N}(0, I)$ is sampled for action selection and held constant for k consecutive steps. The repetition length k is drawn from a Zeta distribution with probability mass function $P(k) \propto k^{-s}$ [12], favoring short repeat intervals while occasionally producing long, correlated action sequences.